

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration, Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 20%

QUESTIONS: 10

Total Marks:20

Annexure A

CLASS TEST

Criteria	Understanding of Task (2)	Procedure & Execution (2.5)	Observation & Result (2)	Viva Voce / Conceptual Response (2)	Time Management & Presentation (1.5)	Total(10)
Weightage	20%	25%	20%	20%	15%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						
<i>Q6</i>						
<i>Q7</i>						
<i>Q8</i>						
<i>Q9</i>						
<i>Q10</i>						

Code of Conduct:

I _____ N.Tejaswitha 192371057 _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

N.Tejaswitha

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Task	20%	Clearly understands the given task/experiment without assistance	Understands task with minor clarification	Partial understanding; requires guidance	Unable to understand the task
Procedure & Execution	25%	Performs procedure accurately and independently with proper technique	Performs with minor errors	Requires guidance; makes procedural mistakes	Unable to execute the procedure
Observation & Result	20%	Records accurate observations and obtains correct results	Minor errors in observation or result	Incomplete or partially correct results	Incorrect or no results
Viva Voce / Conceptual Response	20%	Answers all questions confidently with strong conceptual clarity	Answers most questions with minor gaps	Limited responses; needs prompting	Unable to answer questions
Time Management & Presentation	15%	Completes task on time with neat and clear presentation	Completes with minor delay or presentation issues	Delayed or poorly presented work	Unable to complete on time

CLASS TEST

QUESTIONS:

Total Marks: 10x2 =20

Q1. Define Data Mining and list any two basic data mining tasks. Explain the steps involved in KDD process and justify the role of data preprocessing in it. [L2, L4]

Q2. What is data cleaning? Mention two common data quality issues. Explain methods to handle missing values and noisy data with suitable examples. [L1, L2, L3]

Q3. Define data integration and list two challenges associated with it. Explain how redundancy and inconsistency are handled during data integration. [L2, L4]

Q4. What is covariance analysis? List the two covariance formalisms [L2, L3]

Q5. Suppose two stocks A and B have the following values in one week: (2, 5), (3, 8), (5, 10), (4, 11), (6, 14). Question: If the stocks are affected by the same industry trends, will their prices rise or fall together and independent ?

Q6. Explain the architecture of a typical data mining system with a neat diagram. Discuss the role of each component and analyze how they interact to support knowledge discovery. [L2, L4]

Q7. Find covariance for following data set $x = \{4, 5, 6, 7, 9\}$, $y = \{8, 3, 7, 5, 6\}$

Q8. Describe the major steps in data pre-processing. Given a dataset with missing values, outliers, and inconsistent formats, propose a systematic pre-processing pipeline and justify the sequence of steps. [L3, L4, L5]

Q9. A hospital is analyzing patient **age** data to identify broad health trends. Because the raw data contains minor variations and noise, the analyst needs to perform local smoothing using **equal-frequency** (equi-depth) binning. **Sample Data Set (Ages):** 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 [L2, L4]

Q10. Explain data discretization and its importance in data mining. Compare equal-width and equal-frequency binning with suitable examples, and evaluate their impact on data analysis. [L3, L4, L5]

1. Data Mining : Finds useful and hidden patterns from large datasets.

* Two Data Mining Tasks:

- **classification**: Assigns data into predefined categories.
- **clustering**: Groups similar data objects together.

* KDD steps:

Data Selection



Data cleaning



Data Transformation



Data Mining



Pattern Evaluation



Knowledge Presentation.

* Preprocessing: Removes missing, noisy & inconsistent data.

2. Data cleaning:

- Removes (or) corrects incorrect, incomplete & noisy data.

* Two Data Quality Issues:

- Missing values: some data values are absent.
- Noisy Data: Data contains errors (or) unwanted variations.

* Handling Missing values:

- Use Mean

Median

Mode (or) remove the record.

* Handling Noisy Data:

- Use binning, regression (or) clustering to smooth (or) identify noise

3. *Data Integration: Combines data from different sources into one unified dataset.

* Two challenges:

- Schema Differences: Sources may have different structures (of) attribute names.

- Data Redundancy: The same data may appear multiple times.

* Redundancy: Remove duplicate records (of) attributes.

* Inconsistency: Standardize names, formats, and units to maintain consistent data.

4. *Covariance Analysis: Measures how 2 variables change together

* Positive covariance: Both variables tend to increase (of) decrease together.

* Negative covariance: one variable increases while the other tends to decrease.

* Two covariance formulas:

- Population:

$$\text{Cov}(X, Y) = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{n}$$

- Sample:

$$\text{Cov}(X, Y) = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{n-1}$$

5. Stock Covariance:

Given, stock A = {2, 3, 5, 4, 6}

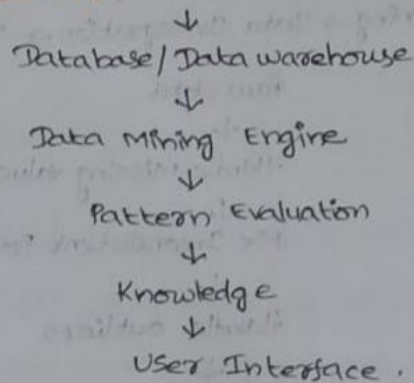
stock B = {5, 8, 10, 11, 14}

* covariance: 5 (Using simple covariance)

* Result: The covariance is positive.

* CONCLUSION: Both stocks have a positive relationship & tend to rise (of) fall together due to similar industry trends rather than behaving independently.

6. *Data Mining System Architecture: Data Sources



* components:

- Data Sources: Provide data
- Data Mining Engine: Finds useful patterns
- Knowledge Base: Provides domain knowledge.
- Pattern Evaluation: selects interesting patterns.
- User Interface: Displays discovered results.

7. * Covariance calculation:

Given; $X = \{4, 5, 6, 7, 9\}$

$Y = \{8, 3, 7, 5, 6\}$

* Mean of X:

6.2

* Mean of Y:

5.8

* Sum of deviation products: -1.8

* Population covariance:

$$\text{Cov}(X, Y) = \frac{-1.8}{5} = -0.36$$

* Sample Covariance:

$$\text{Cov}(X, Y) = \frac{-1.8}{4} = -0.45$$

* CONCLUSION:

→ Negative covariance shows an inverse relationship between X and Y.

8. * Data Pre-processing Major steps:

Data cleaning → Data Integration → Data Transformation → Data Reduction.

* Pipeline:

Raw data



Handle Missing values



Fix Inconsistent Formats



Handle outliers



Transform data...

* Result: Produces clean & consistent data for accurate data Mining

9. Equal-Frequency Binning:

* Dataset: contains 27 age values.

* Bin 1: 13, 15, 16, 16, 17, 20, 20, 21, 22

⇒ Mean = 18

* Bin 2: 22, 25, 25, 25, 25, 30, 33, 33, 35

⇒ Mean = 28.11

* Bin 3: 35, 35, 35, 36, 40, 45, 46, 52, 70

⇒ Mean = 43.78

* Result: {18, 18, 18, 18, 18, 18, 18, 18, 28.11, 28.11, 28.11, 28.11, 28.11, 28.11, 28.11, 28.11, 28.11, 43.78, 43.78, 43.78, 43.78, 43.78, 43.78, 43.78, 43.78, 43.78, 43.78}

10. * Data Discretization:

⇒ converts continuous data into intervals (or) categories.

* Importance:

- simplifies data
- Reduces Noise
- Makes patterns easier to understand.

* Equal-Width: Divides the range into equal-sized intervals.

* Equal-Frequency: divides data into bins with nearly equal numbers of values.

* Conclusion: Equal-width is useful for fixed ranges, while Equal-frequency is better than uneven data.